



SOLIDWORKS STAPLER RE-ENGINEER PROJECT

By Gabriel Ragonjan

Tactile Orientation Feature

Design Problem:

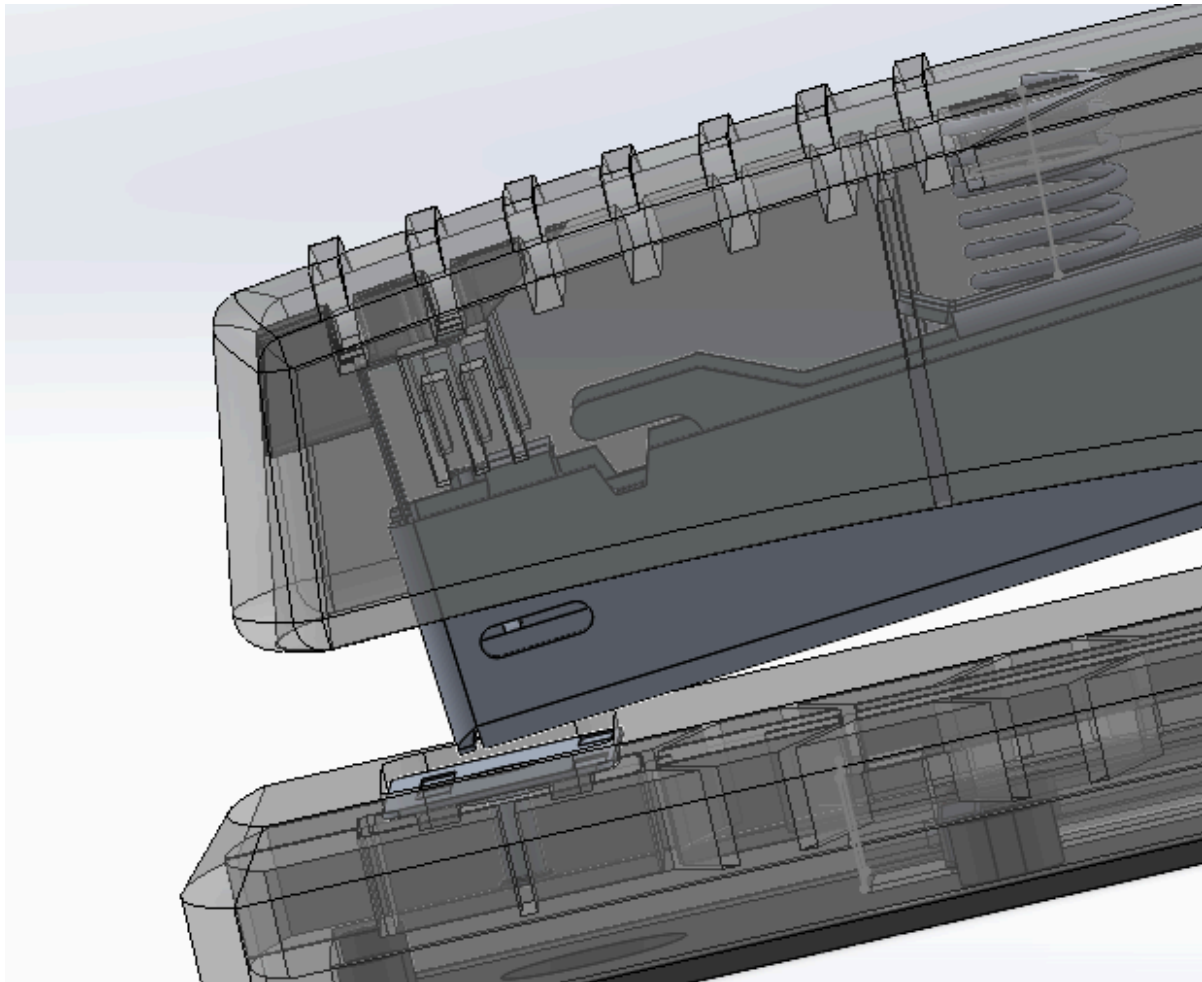
A visually impaired user may have difficulty determining the front and orientation of the stapler through touch alone.

Design Modification:

Added a textured surface to the front of the stapler to provide a tactile reference point.

Engineering Purpose:

Allows the user to identify the front of the stapler through touch without significantly changing the original geometry.



Integrated Staple Storage

Design Problem:

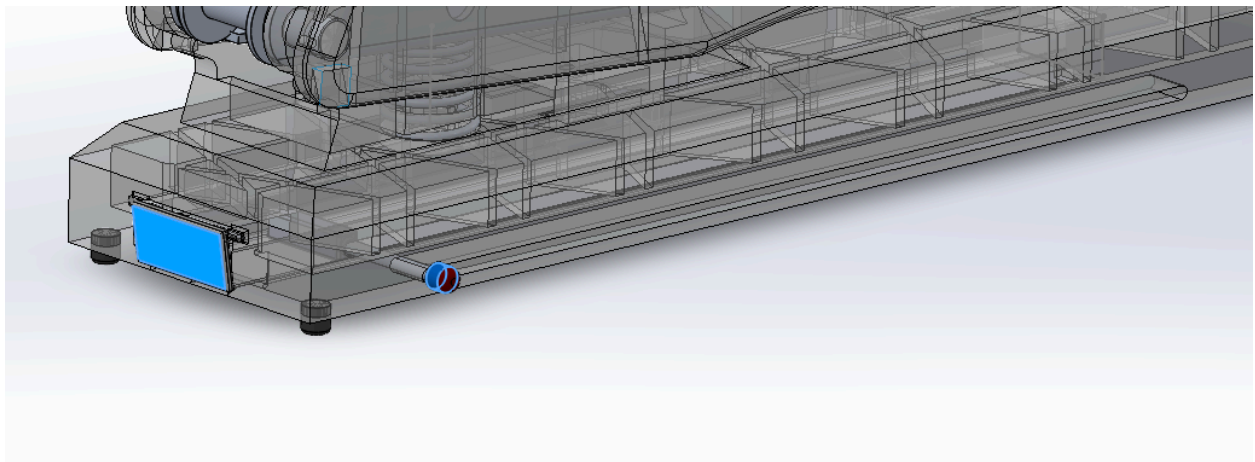
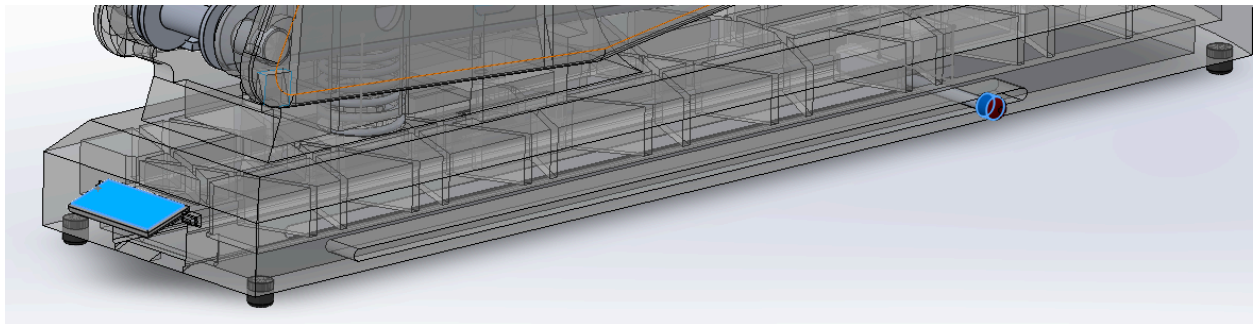
The original stapler provides no integrated storage for additional staples.

Design Modification:

Designed an internal storage compartment with a sliding bolt/dispenser mechanism and hinged access door.

Engineering Purpose:

Adds staple storage while keeping the mechanism contained within the existing base.



Redesigned Base Support

Design Problem:

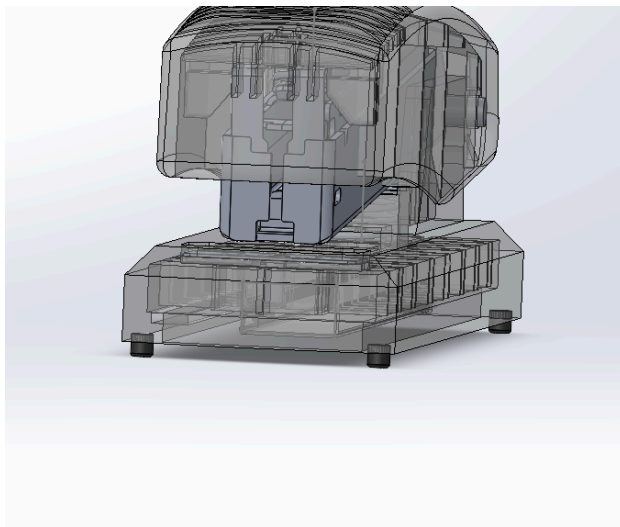
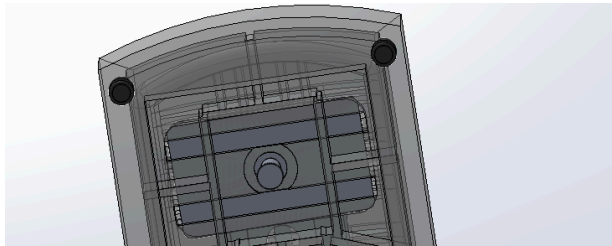
The original design used a large rubber pad covering most of the stapler base, requiring more material than necessary to provide grip and stability.

Design Modification:

Replaced the large rubber pad with smaller, strategically positioned rubber feet while retaining the original mounting features.

Engineering Purpose:

The redesign aimed to maintain adequate grip and stability while reducing the amount of rubber required, potentially lowering material and manufacturing costs. Using smaller individual feet also concentrates the high-friction material at the points where it is actually needed.



FINAL DESIGN

The completed redesign combines tactile orientation, integrated staple storage, and revised base support into the original stapler architecture.

Software: SolidWorks

Skills: 3D CAD · Mechanical Design · Design Iteration · Product Redesign · Accessibility Design · Reverse Engineering